

System Architecture Design

Allowing users to stake ERC20 tokens and earn rewards

1. Solidity
2. React
3. Typescript
4. Ether.js
5. Metamask

Features:

- Stake tokens
- Withdraw tokens
- Claim rewards
- View dashboard



System Architecture

The DeFi Staking Platform consists of five main components:

1. User – interacts with the application through the frontend.
2. MetaMask Wallet – manages user accounts and signs blockchain transactions.
3. React Frontend – provides the user interface for staking, withdrawing, and claiming rewards.
4. Staking Smart Contract – handles staking operations, reward calculations, and withdrawals.
5. ERC20 Token Contract – manages token balances and transfers.

The frontend communicates with the staking contract using Ethers.js. Users connect their wallets through MetaMask and interact with the staking contract to stake tokens, claim rewards, and withdraw funds.